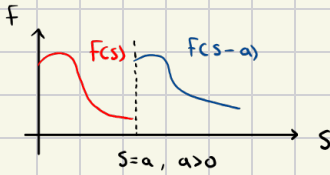


Operational Properties

translation on the s-axis



Shift on the s-axis

7.3.1 First translation theorem

if $\mathcal{L}\{f(t)\} = f(s)$ and a all real numbers then:

$$\mathcal{L}\{e^{at} f(t)\} = f(s-a)$$

ex:

$$\mathcal{L}\{e^{st} t^3\} \Rightarrow \mathcal{L}\{t^3\}_{s \rightarrow s-s} \Rightarrow \frac{3!}{s^4} = \boxed{\frac{6}{(s-s)^4}}$$

$t^n = \frac{n!}{s^{n+1}}$

all we did was apply the Laplace theorem on t^3 and look
 $e^{at} = (s-a)$ then plugged that into "s"

$$y'' - 4y' + 4y = t^3 e^{2t}, \quad y(0) = 0, \quad y'(0) = 0$$

$$\mathcal{L}\{y''\} - 4\mathcal{L}\{y'\} + 4\mathcal{L}\{y\} = \mathcal{L}\{t^3 e^{2t}\}$$

$$s^2 y(s) - sy(0) - y'(0) - 4[sy(s) - y(0)] + 4y(s) = \frac{6}{(s-2)^4}$$

$$s^2 y(s) - \cancel{sy(0)} - \cancel{y'(0)} - 4sy(s) + \cancel{4y(0)} + 4y(s) = \frac{6}{(s-2)^4}$$

$$y(s) (s^2 - 4s + 4) = \frac{6}{(s-2)^4}$$

$$y(s) (s-2)^2 = \frac{6}{(s-2)^4}$$

$$y(s) = \frac{6}{(s-2)^6}$$

notice $t^n = \frac{n!}{s^{n+1}}$ the numerator should be the denominator minus one for this problem $6-1=5$, so 5 should be the numerator

so:

$$\frac{1}{20} \frac{5!}{(s-2)^6}$$

(n-1)

now we got the numerator as $5! = 120$ if we divide 120 by 20 we get 6

$\Rightarrow$

$$y(t) = \frac{1}{20} t^5 e^{2t}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 2s + 5} \right\}$$

- notice we cannot break the denominator easily so we would need to complete the square

$$s^2 + 2s + 1 = -s + 1$$

$$(s+1)^2 = -4$$

$$(s+1)^2 + 4 = 0$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2 + 4} \right\}$$

- we notice that this will either be $\sin(kt)$ or $\cos(kt)$ we know it isn't going to be $\cos(kt)$ because the numerator isn't "s" so its going to be $\sin(kt)$ so we need to have "k" in numerator

$$\mathcal{L}^{-1} \left\{ \frac{1}{2} \cdot \frac{2}{(s+1)^2 + 2^2} \right\}$$

$$\Rightarrow \boxed{\frac{1}{2} e^{-t} \sin(2t)}$$

$$y'' - 6y' + 9y = t, \quad y(0) = 0, \quad y'(0) = 1$$

$$\mathcal{L}\{y''\} - 6\mathcal{L}\{y'\} + 9\mathcal{L}\{y\} = \mathcal{L}\{t\}$$

$$s^2 y(s) - sy(0) - y'(0) - 6[sy(s) - y(0)] + 9y(s) = \frac{1}{s^2}$$

$$s^2 y(s) - \cancel{0} - (1) - 6sy(s) + \cancel{0} + 9y(s) = \frac{1}{s^2}$$

$$s^2 y(s) - 6sy(s) + 9y(s) = \frac{1}{s^2} + 1$$

$$y(s) (s^2 - 6s + 9) = \frac{1+s^2}{s^2}$$

$$y(s) (s-3)^2 = \frac{1+s^2}{s^2}$$

$$y(s) = \frac{1+s^2}{s^2 (s-3)^2}$$

• You need to apply Partial decomposition

$$\left[\frac{A}{s} + \frac{B}{s^2} + \frac{C}{(s-3)} + \frac{d}{(s-3)^2} = \frac{1+s^2}{s^2 (s-3)^2} \right] s^2 (s-3)^2$$

$$As(s-3)^2 + B(s-3)^2 + Cs^2(s-3) + ds^2 = 1+s^2$$

$$As(s^2-6s+9) + B(s^2-6s+9) + Cs^2(s-3) + ds^2 = 1+s^2$$

$$\underline{As^3} - \underline{6As^2} + 9As + \underline{Bs^2} - 6Bs + 9B + \underline{Cs^3} - \underline{3Cs^2} + \underline{ds^2} = 1+s^2$$

$$As^3 - Cs^3 = 0$$

$$-6As^2 - Bs^2 + Cs^2 + ds^2 = s^2$$

$$9As - 6Bs = 0$$

$$9B = 1$$

$$A - C = 0$$

$$-6A - B + C + d = 1$$

$$9A - 6B = 0$$

$$B = \frac{1}{9}$$

$$-6A - B + C + d = 1$$

$$-6\left(\frac{2}{27}\right) - \left(\frac{1}{9}\right) + \left(\frac{2}{27}\right) + d = 1$$

$$\frac{-13}{27} + d = 1$$

$$d = \frac{40}{27}$$

$$9A - 6B = 0$$

$$9A - 6\left(\frac{1}{9}\right) = 0$$

$$9A - \frac{2}{3} = 0$$

$$9A = \frac{2}{3}$$

$$A = \frac{2}{27}$$

$$A - C = 0$$

$$A = C$$

$$C = \frac{2}{27}$$

$$Y(s) = \left\{ \frac{2}{27} \frac{1}{s} + \frac{1}{9} \frac{1}{s^2} + \frac{2}{27} \frac{1}{(s-3)} + \frac{40}{27} \frac{1}{(s-3)^2} \right\}$$

(n-1)

$$\frac{2}{27} + \frac{1}{9}t + \frac{2}{27}e^{3t} + \frac{40}{27}te^{3t}$$

$$\mathcal{L}\{(1 - e^t + 4e^{-2t}) \cos(6t)\}$$

• you need to distribute the $\cos(6t)$ into the Parathesis

$$\mathcal{L}\{(\cos(6t) - e^t \cos(6t) + 4e^{-2t} \cos(6t))\}$$

$$\mathcal{L}\{\cos(6t)\} - \mathcal{L}\{e^t \cos(6t)\} + 4\mathcal{L}\{e^{-2t} \cos(6t)\}$$

$$e^{at} = (s-a)$$

$$\frac{s}{s^2 + 36} - \left\{ \frac{s}{s^2 + 36} \right\}_{s \rightarrow s-1} + 4 \left\{ \frac{s}{s^2 + 36} \right\}_{s \rightarrow s+2}$$

$$\boxed{\frac{s}{s^2 + 36} - \frac{(s-1)}{(s-1)^2 + 36} + \frac{4(s+2)}{(s+2)^2 + 36}}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 8s + 17}\right\}$$

• you would need to apply complete the square on this problem because the denominator cannot be broken into two factors

$$s^2 + 8s + 16 = -17 + 16$$

$$(s+4)^2 - 1 = 0$$

• When you see a problem like this that has a "s" in the numerator you need to break it into two by doing "s+4 -4" use what ever number you have

$$\mathcal{L}^{-1}\left\{\frac{s}{(s+4)^2 - 1}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s+4-4}{(s+4)^2 - 1}\right\} \rightarrow \mathcal{L}^{-1}\left\{\frac{(s+4)}{(s+4)^2 - 1} - \frac{4}{(s+4)^2 - 1}\right\}$$

$$\mathcal{L}^{-1} \left\{ \frac{(s+4)}{(s+4)^2 - 1} - 4 \frac{1}{(s+4)^2 - 1} \right\}$$

• this has $(s+4)$
in numerator and
 $(s+4)^2$ in denominator
so we can tell that
this is $\cos(4t)$

• this has one in
the numerator the "u"s
match so this is $\sin(4t)$

$$e^{-4t} (\cos(4t) - 4 \sin(4t))$$

$$y'' + 2y' + y = 0, \quad y(0) = 1, \quad y'(0) = 1$$

$$\mathcal{L}\{y''\} + 2\mathcal{L}\{y'\} + \mathcal{L}\{y\} = 0$$

$$s^2 y(s) - sy(0) - y'(0) + 2[sy(s) - y(0)] + y(s) = 0$$

$$s^2 y(s) - s(1) - (1) + 2sy(s) - 2(1) + y(s) = 0$$

$$s^2 y(s) - s - 3 + 2sy(s) + y(s) = 0$$

$$s^2 y(s) + 2sy(s) + y(s) = s + 3$$

$$y(s) (s^2 + 2s + 1) = s + 3$$

$$y(s) (s+1)^2 = s + 3$$

$$y(s) = \frac{s+3}{(s+1)^2}$$

$$y(s) = \frac{s+1+2}{(s+1)^2}$$

$$y(s) = \frac{(s+1)}{(s+1)^2} + \frac{2}{(s+1)^2}$$

$$y(s) = \frac{1}{(s+1)} + 2 \cdot \frac{1}{(s+1)^2}$$

$$y(s) = \frac{1}{(s+1)} + 2 \cdot \frac{1}{(s+1)^2}$$

$$e^{-t} + 2te^{-t}$$
